2.5 Pre-Quiz: Trigonometry and Statistics — Markscheme

Total: 47 marks. Accuracy marks need 3 s.f., the exact value, or the calculator display with an ellipsis. In Problem 3 the calculator must be in radian mode: a value worked in degree mode earns the M mark if the method is right, and A0.

1. [Maximum mark: 7]

(a) $\tan 41^\circ = \dfrac{FT}{60}$ **M1**

$FT = 60\tan 41^\circ = 52.2\,\text{m} \quad (52.1572\ldots)$ **A1**

Wrong methods: $60\sin 41^\circ = 39.4$ *or* $\frac{60}{\tan 41^\circ} = 69.0$ *earn M0A0.*

(b) $BF = \dfrac{52.1572\ldots}{\tan 26^\circ}$ **M1**

$BF = 106.938\ldots$ **A1**

$AB = 106.938\ldots - 60 = 46.9\,\text{m} \quad (46.9381\ldots)$ **A1**

Accept the route through triangle ABT: $A\hat{T}B = 15^\circ$, $AT = \frac{60}{\cos 41^\circ} = 79.50\ldots$, *then the sine rule. Giving* $BF = 107$ *as the answer for AB earns M1A1A0. Accept* 47.0 *from the rounded height* 52.2. *FT from (a).*

(c) $BT = \dfrac{52.1572\ldots}{\sin 26^\circ}$ **M1**

$BT = 119\,\text{m} \quad (118.979\ldots)$ **A1**

Accept $\frac{BF}{\cos 26^\circ}$ *or Pythagoras with BF and FT. Wrong method:* $\frac{FT}{\cos 26^\circ} = 58.0$ *earns M0A0. FT from (a) or (b).*

Feedback: carry the full height forward; round only the final answer.

2. [Maximum mark: 7]

(a) $AB^2 = 14^2 + 9^2 - 2(14)(9)\cos 118°$ **M1A1**

$AB^2 = 395.306\ldots$

$AB = 19.9\,\text{km}$ $\quad (19.8823\ldots)$ **A1**

M1 for the cosine rule, A1 for the correct substitution. Wrong method: adding $2(14)(9)\cos 118°$ *gives* 12.6*: M1A0A0. The answer must be greater than* 14*, the longest given side, because* $118°$ *is the largest angle.*

(b) $\dfrac{\sin H\hat{A}B}{9} = \dfrac{\sin 118°}{19.8823\ldots}$ **M1**

$H\hat{A}B = 23.6°$ $\quad (23.5580\ldots)$ **A1**

M1 for pairing $HB = 9$ *with* $H\hat{A}B$*. Wrong method: pairing* 14 *with* $H\hat{A}B$ *gives* $38.4°$*, which is* $H\hat{B}A$*: M0A0. Accept the cosine rule. Accept* 23.5 *from the rounded* $AB = 19.9$*. FT from (a).*

(c) Area $= \frac{1}{2}(14)(9)\sin 118°$ **M1**

Area $= 55.6\,\text{km}^2$ $\quad (55.6256\ldots)$ **A1**

Accept any two sides with the angle between them. Wrong method: $\frac{1}{2}(14)(9) = 63$ *earns M0A0.*

3. [Maximum mark: 7]

(a) $10^2 = x^2 + (2x)^2 - 2(x)(2x)\cos 1.2$ **M1A1**

$100 = x^2\,(5 - 4\cos 1.2) = 3.55056\ldots x^2$

$x = 5.31 \quad (5.30702\ldots)$ **A1**

M1 for the cosine rule with $BC = 10$ opposite the angle 1.2; A1 for the correct substitution. Wrong methods: writing $(2x)^2$ as $2x^2$ gives $x = 8.03$ (M1A0A0); degree mode gives $x = 10.0$ (M1A1A0).

(b) Area $= \frac{1}{2}(x)(2x)\sin 1.2 = (5.30702\ldots)^2 \sin 1.2$ **M1**

Area $= 26.3\,\text{cm}^2 \quad (26.2504\ldots)$ **A1**

Degree mode gives 0.590: M1A0. FT from (a).

(c) $\dfrac{\sin A\hat{C}B}{5.30702\ldots} = \dfrac{\sin 1.2}{10}$ **M1**

$A\hat{C}B = 0.517 \quad (0.517415\ldots)$ **A1**

M1 for pairing $AB = x$ with $A\hat{C}B$. Wrong method: pairing $2x$ with $A\hat{C}B$ gives 1.42, which is $A\hat{B}C$: M0A0. The answer must be in radians: $29.6°$ earns M1A0. Accept 0.518 from the rounded $x = 5.31$. FT from (a).

Feedback: store x in the calculator and use the stored value.

4. [Maximum mark: 8]

(a) $A\hat{C}B = 180° - 72° - 51° = 57°$ **A1**

(b) $\dfrac{AC}{\sin 51°} = \dfrac{45}{\sin 57°}$ **M1**

$AC = 41.7\,\text{m}$ $\quad (41.6988\ldots)$ **A1**

M1 for pairing AC with $A\hat{B}C = 51°$ and $AB = 45$ with $A\hat{C}B$. Wrong method: pairing AC with $72°$ gives 51.0, which is BC: M0A0. FT from (a).

(c) Area $= \frac{1}{2}(45)(41.6988\ldots)\sin 72°$ **M1**

Area $= 892\,\text{m}^2$ $\quad (892.303\ldots)$ **A1**

M1 for two sides with the angle between them. Wrong method: using $\sin 57°$ with AB and AC gives 787: M0A0. FT from (b).

(d) $\frac{1}{2}(30)(40)\sin\theta = 500$ **M1**

$\sin\theta = \frac{5}{6}$, $\quad$ acute value $56.4°$ **(A1)**

$\theta = 180° - 56.4426\ldots° = 124°$ $\quad (123.557\ldots)$ **A1**

Giving only the acute value $56.4°$ earns M1A1A0. A1 for 124 (3 s.f.) or the display with an ellipsis; 123.6 with no ellipsis earns A0.

5. [Maximum mark: 10]

Ordered: $28, 31, 33, 34, 36, 37, 38, 39, 40, 41, 43, 45, 54, 85$ ($n = 14$, sum 584).

With $n = 14$ even, each half has seven values, so the GDC and the hand method give the same quartiles.

(a) Median $= \frac{38+39}{2} = 38.5$ **A1**

$Q_1 = 34, \quad Q_3 = 43$ **A1**

(b) IQR $= 43 - 34 = 9$ **A1**

upper fence $= Q_3 + 1.5 \times \text{IQR} = 43 + 13.5$ **M1**

$= 56.5$ **A1**

$54 < 56.5$, so 54 is not an outlier; $85 > 56.5$, so 85 is an outlier **R1**

R1 for both conclusions with the comparison stated. The lower fence, $34 - 13.5 = 20.5$, is not required. Wrong methods: adding $1.5 \times IQR$ to the median ($38.5 + 13.5 = 52$), or adding the IQR without the 1.5 ($43 + 9 = 52$), makes 54 an outlier: M0A0R0. FT from (a).

(c) $\bar{x} = 41.7 \quad (41.7142\ldots)$ **A1**

$\sigma = 13.5 \quad (13.5088\ldots)$ **A1**

Do not accept $s_x = 14.0$ ($14.0188\ldots$), the sample standard deviation.

(d) $\dfrac{584 - 85}{13}$ **M1**

$= 38.4 \quad (38.3846\ldots)$ **A1**

Accept re-entering the 13 values in the GDC. Wrong methods: dividing by 14 (35.6), or giving the new median 38: M0A0.

6. [Maximum mark: 8]

GDC: *enter x and y in two lists and run* `Linear Regression (mx+b)`.

(a) $r = 0.991 \quad (0.990632\ldots)$ **A1**

strong, positive (linear) correlation **A1A1**

A1 for "positive", A1 for "strong" (accept "very strong"), independently. Do not accept $r^2 = 0.981$. *FT from the candidate's* r.

(b) $y = 3.15x + 44.7$ **A1A1**

A1 for $a = 3.15$ *(*$3.15217\ldots$*), A1 for* $b = 44.7$ *(*$44.6847\ldots$*), independently. The reversed line* $x = 0.311y - 13.8$ *earns A0A0.*

(c) $y = 3.15(10) + 44.7$ **(M1)**

$y = 76.2 \quad (76.2065\ldots)$ **A1**

The full-precision line and the rounded line both give 76.2. *Wrong method: rearranging the x-on-y line gives* 76.4*: M0A0. FT from (b).*

(d) 25 hours is outside the range of the data (2 to 12 hours), so the estimate would be an extrapolation **R1**

Accept a reason that the linear pattern may not continue beyond the data, e.g. the line predicts a score of about 123*, above the maximum. Do not accept "r is not strong enough".*